hw12

Please make sure you're following all the instructions described in the "Homework Instructions" page on Canvas (go to the homepage of our class and you'll see a link saying "Homework Instructions" just underneath the header of the page).

Note on Homework 12:

Autograding isn't particularly conducive to plots. As such, the only questions that will make use of the autograder are those that produce some output that the autograder can check. The other questions will show up as having a "out of" score of 0 - these will be graded manually.

Also, for *all* plots, provide useful titles and axis labels.

1 Baby names

We've used the `babynames` package few times in this course - let's now create some plots to explore a few names.

```
bn <- babynames::babynames
```

1.1 *Subset the data*

Let's explore a name with multiple spellings - we'll focus on "Catherine", "Katherine", and "Kathryn".

Subset the data frame to those three names (be sure to also limit it to only females).

var: `bn_sub`

1.2 *Plot the data*

Now create a plot showing how the proportion of each of the three names has changed over time (the plot should have three separate lines, with each line being a different color).

1.3 *Most common names overall*

Let's explore the 20 names that occur most frequently in the data. Create a data frame that has two columns - `name` and `n`, where `n` contains the total number of babies born with that name across all years. The data frame should have 20 rows and should be sorted in descending order.

Hint: this is a multi-step process. `aggregate()` *and* `order()` *will be useful here.*

var: `top_names`

1.4 *Create a bar plot*

Using `top_names`, create a bar plot where the y axis is the total number of babies with that name and the x axis is the baby name. *(Which function is more appropriate here:* `geom_bar()` *or* `geom_col()`*?)*

1.5 *Rotate the axis labels*

You'll notice that the axis labels are overlapping (unless you make the plot area very wide). Let's rotate the axis labels to make them readable. Do some web searching to figure out how to do this and then create a new plot where the names are more readable.

1.6 *Re-order the factors*

You may notice that the names are ordered alphabetically rather than in descending order. To control the order of the bars, we can turn `name` into a factor and then set the order of the levels - the bars will be plotted in that order. Create a copy of `top_names` called `top_names2`. Modify the `name` column to convert it to a factor using the `factor()` function. Note that the `factor()` function can be used as follows:

```r
x <- c("a", "b", "c", "b", "a")
factor(x, levels = c("b", "c", "a"))
```

The `levels` argument can be used to specify the order of the factor - you'll want this second argument to be the names in order from most frequent to least frequent (which we've already calculated).

var: `top_names2` - same as `top_names` but with the `name` column as a factor with the levels ordered as described above

1.7 *Re-create the plot*

Now use `top_names2` to create a bar plot where the names are ordered by their frequency.

2 Hurricanes

Exercise 3.5.8 from "Modern Data Science with R" (2nd Edition) by Baumer, Kaplan, and Horton

The `storms` data from the `nasaweather` package contains tropical cyclone tracks through the Atlantic Ocean, Caribbean Sea, and Gulf of Mexico from 1995 to 2000. Use the `geom_path` function to plot the path of each topical storm. Use color to distinguish the storms from one another, and use faceting to plot each year in its own panel.

Note: you'll notice that the legend gets very cluttered and isn't very useful. You may want to remove the legend - do some web searching to find out how to do this.

3 Trail usage

For this exercise we'll use the `RailTrail` dataset from the `mosaicData` package (which you will likely need to install). First, read the documentation about this dataset. Then explore the data - focus on examining which variables appear to be associated with the `volume` column. At minimum, create one plot of each of the following types:

- Histogram
- Scatter plot with the color or shape determined by another variable
- Box plot
- A plot that uses facets

Write a few sentences describing any associations revealed by your plots.